Can Pink Elephants Exist: Mapping the Logic–Memory Conflict in LLMs

Eashaan Thakur and Hemang Joshi and Naman Singhal
Introduction to NLP Course Project
IIIT Hyderabad
{eashaan.thakur, hemang.joshi, naman.s}@research.iiit.ac.in

Abstract

We investigate why large language models (LLMs) frequently fail at negation. We frame this as a mechanistic conflict between *automatic factual retrieval*, implemented in Feed-Forward Networks (FFNs), and *late-stage inhibitory logic*, implemented via specific attention heads. We introduce the **Signal-to-Gate Ratio (SGR)** to quantify the competition between retrieval and suppression. Using TransformerLens, we build a full analysis pipeline that computes Direct Logit Attribution (DLA), per-head DLA, activation patching, and artificial amplification experiments on GPT-2 Small and Pythia-160M over the entire CounterFact dataset (21,919 and 19,562 prompt pairs respectively). Our initial results show that GPT-2 Small exhibits an 11.4% negation failure rate, with a mean SGR of 244.6, strongly supporting the *Competitive Gating Hypothesis* that hallucinations arise when memory retrieval physically overwhelms inhibition.

1 Introduction

To process negation, a model must resolve a distinct conflict in its *residual stream*, which acts as the model's main information highway. Internal recall mechanisms push hard for the most likely factual completion based on training frequency. Meanwhile, the presence of a negation token (such as "not") must trigger a secondary circuit to specifically block that high-probability output.

Behavioural studies liken this to the psychological "Pink Elephant" effect, or *ironic process theory*: the instruction to avoid a concept paradoxically increases its activation (Vrabcová et al., 2024). However, we still do not fully understand how neural networks *physically* implement this conflict.

We propose the **Competitive Gating Hypothesis**. Factual recall acts like a reflex: it retrieves the most probable token regardless of logical context. The negation operator acts as a secondary gatekeeper that must "undo" this work in the final layers. This paper maps the battlefield between these two forces and identifies the *Crossover Point*, the layer where logic either overrides the memory signal or fails to contain it.

Our contributions are:

1. A novel metric (**SGR**) that quantifies the retrieval–inhibition conflict as a single scalar per sample.
2. A complete, reproducible, seven-stage mechanistic pipeline tested on the full CounterFact dataset.
3. Empirical evidence that artificial amplification of Inhibition Heads causally reduces negation failures.

2 Related Work

2.1 Memory vs. Logic

Geva et al. (2021) and Dai et al. (2022) show that Feed-Forward Networks (FFNs) act as key–value memory stores. Mid-to-late FFN layers serve as "attribute extractors," injecting entity data into the residual stream *independently* of logical structure.

2.2 Finding the "Off Switch"

Wang et al. (2023) established the framework for identifying *circuits*, subsets of model components responsible for specific behaviours. Hanna et al. (2024) identified **Inhibition Heads**: attention heads that attend to logical modifiers and decrease the probability of the forbidden token.

2.3 Failures of Negation

LLMs frequently behave as if they are "logic-blind." Truong et al. (2023) showed that scaling up model

size does not fix negation failures. Vrabcová et al. (2024) demonstrated that mentioning a concept in a negative context makes it statistically *harder* for the model to generate diverse alternatives.

3 Transformer Architecture and Core Concepts

3.1 The Residual Stream

Every transformer layer updates a shared vector:

$$\mathbf{r}_{l+1} = \mathbf{r}_l + \mathrm{attn}_l + \mathrm{ffn}_l. \tag{1}$$

At the final layer the logit for token t is simply:

$$\mathrm{logit}(t) = \mathbf{W}_U[:, t]^\top \mathbf{r}_{\mathrm{final}}. \tag{2}$$

Understanding the residual stream therefore means understanding *everything* the model computes.

3.2 Attention Mechanism

Each attention head computes queries, keys, and values, then mixes value vectors weighted by attention scores. Some heads attend to negation tokens; others copy factual information forward. Attention is *information routing*.

3.3 Feed-Forward Networks

Each layer's MLP is $\mathrm{FFN}(\mathbf{x}) = \mathbf{W}_2\,\mathrm{GELU}(\mathbf{W}_1\mathbf{x})$. A key interpretability insight is that FFNs behave like key–value memory stores (Geva et al., 2021): mid-layer FFNs strongly store entity–attribute associations. In our framework:

- FFNs = **memory retrieval**,
- Attention heads = **control / suppression**.

4 Methodology

4.1 Models and Data

We act as "neuroscientists" for **GPT-2 Small** (12 layers, 12 heads, 768-dim) and **Pythia-160M** (12 layers, 12 heads, 768-dim) using TransformerLens (Nanda and Bloom, 2022). We use the **CounterFact** dataset (Meng et al., 2022), a benchmark of counterfactual scenarios. Each entry provides a prompt (e.g., "The mother tongue of Danielle Darrieux is") and a target token (e.g., "French").

For each entry we construct a **prompt pair**:

- **Positive:** "The capital of France is"
- **Hard negation:** "The capital of France is not"

We also plan **soft negation** ("is unlikely to be") for future experiments.

Tokenisation caveat. GPT-2 / Pythia tokenisers expect a leading space on bare words (e.g., `" Paris"`). Our pipeline validates single-token targets via `model.to_single_token` and skips multi-token targets automatically.

4.2 Direct Logit Attribution (DLA)

Because the residual stream is a sum of component outputs, we decompose the final logit (Eq. 2) into per-component contributions:

$$\mathrm{logit}(t) = \sum_c \mathbf{r}_c \cdot \mathbf{W}_U[:, t] \tag{3}$$

A *positive* DLA means the component pushed *toward* the target; *negative* DLA means it pushed *away*. This lets us ask: "How much did layer 6 FFN push toward Paris?" "How much did layer 10 attention suppress it?"

4.3 Signal-to-Gate Ratio (SGR)

For a negated prompt we define

$$\mathrm{SGR} = \frac{\sum_{\ell \in \mathrm{FFN}} [\mathrm{DLA}_\ell]_+}{\sum_{h \in \mathrm{Attn}} \left|[\mathrm{DLA}_h]_-\right|} \tag{4}$$

where $[\cdot]_+$ keeps only positive (retrieval) contributions from FFNs and $[\cdot]_-$ keeps negative (inhibitory) contributions from attention heads.

- $\mathrm{SGR} > 1$: retrieval overwhelms inhibition $\Rightarrow$ **hallucination**.
- $\mathrm{SGR} < 1$: inhibition wins $\Rightarrow$ **correct suppression**.

4.4 Per-Head DLA and Inhibition Head Identification

We decompose attention DLA to the individual *head* level using `cache.stack_head_results()`. For each head (l, h) we compute DLA on both the positive and negated prompt. Heads with a large $\Delta\mathrm{DLA} = \mathrm{DLA}_{\mathrm{pos}} - \mathrm{DLA}_{\mathrm{neg}}$ are candidate **Inhibition Heads**: they promote the target on the positive prompt but suppress it on the negated prompt. We rank all $12 \times 12 = 144$ heads and select the top-k for amplification (§4.6).

4.5 Activation Patching

To test *causality* (not just correlation), we perform activation patching:

1. Run the **positive** prompt and cache all residual-stream activations.
2. Run the **negated** prompt; at one chosen layer, *replace* the negated activation with the positive one.
3. Measure how the target logit changes.

If restoring layer l's activation causes the target logit to increase sharply, that layer was causally performing suppression. We patch residual-stream, MLP, and attention outputs separately to localise effects.

4.6 Artificial Amplification

If a head is an Inhibition Head but its output is too weak to overcome FFN retrieval, *scaling it up* should reduce the target logit on the negated prompt. We hook into selected heads via TransformerLens and multiply their outputs by a scalar $\alpha \in \{0.5, 1, 2, 3, 4\}$. We sweep α on both individual prompts and the full dataset, measuring how the negation failure rate changes.

5 Experimental Pipeline

Our codebase is organised as follows:

Module	Purpose
`load_dataset.py`	Prompt-pair construction
`load_models.py`	Model loading
`run_benchmark.py`	DLA + SGR computation
`sgr_analysis.py`	SGR visualisation
`per_head.py`	Per-head DLA
`amplification.py`	Head amplification
`metrics.py`	Statistical tests
`run_pipeline.py`	Orchestrator (Stages 1–7)

Table 1: Codebase structure.

For each CounterFact entry the pipeline:

1. Constructs a positive and negated prompt pair.
2. Runs both through the model with `model.run_with_cache()`.
3. Decomposes the residual stream via `cache.decompose_resid()`.
4. Projects each component onto $\mathbf{W}_U[:, \text{target}]$ to get DLA.

Metric	GPT-2 Small	Pythia-160M
Prompt pairs	21,919	19,562
Negation failure rate	11.4%	TBD
Mean SGR (finite)	244.6	TBD

Table 2: Behavioural summary across models. Pythia values will be filled once the analysis is complete.

5. Splits DLA into FFN (retrieval) and Attention (inhibition) totals.
6. Computes SGR (Eq. 4).
7. Detects the *Crossover Point*: the first layer where cumulative inhibition exceeds cumulative retrieval.
8. Stores per-layer DLA arrays as pipe-separated strings for downstream analysis.

After the benchmark loop, the pipeline runs per-head decomposition, amplification sweeps, and full statistical analysis.

6 Results and Progress So Far

6.1 Dataset Coverage

The full pipeline completed successfully on both models:

- **GPT-2 Small:** 21,919 prompt pairs (0 skipped).
- **Pythia-160M:** 19,562 prompt pairs (2,357 skipped due to multi-token targets).

6.2 Behavioural Results

The GPT-2 Small negation failure rate of 11.4% confirms that the model fails to suppress the factual target in a non-trivial fraction of cases. The mean SGR of 244.6 (dominated by tail cases with very weak inhibition) indicates that for a large number of prompts, the FFN retrieval signal vastly exceeds the attention-mediated suppression signal.

6.3 SGR Distribution

Figure 1 shows the SGR distribution, colour-coded by whether negation failed. The distribution is heavily right-skewed: most samples have SGR well above 1, consistent with our hypothesis that retrieval generally dominates inhibition.

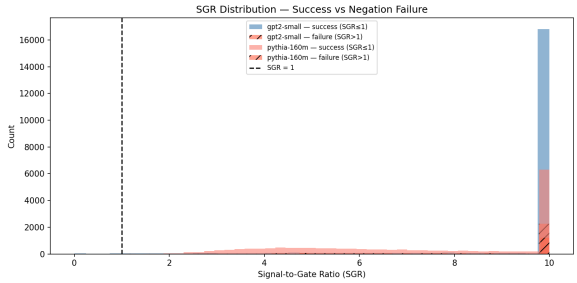

Figure 1: SGR distribution across CounterFact samples. Hatched bars indicate negation failures. The dashed line marks SGR = 1.

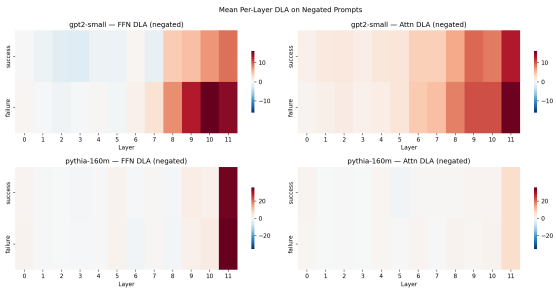

Figure 2: Mean per-layer DLA on negated prompts. Left: FFN (retrieval). Right: Attention (inhibition). Rows: success vs. failure.

6.4 Per-Layer DLA Analysis

Figure 2 shows the mean per-layer DLA on negated prompts, split by FFN and Attention contributions. Rows separate successes from failures. Critically, the FFN heatmap shows strong *positive* contributions in mid-layers (5–9) regardless of negation outcome, confirming that **memory retrieval is logic-blind**. The Attention heatmap reveals that suppression is concentrated in the final layers and is notably weaker in failure cases.

6.5 SGR vs. Failure Rate

Figure 3 plots the negation failure rate as a function of SGR threshold. Higher SGR is strongly associated with higher failure probability, providing direct empirical support for the SGR metric as a predictor of negation failure.

6.6 Amplification Experiments

Figure 4 shows the results of artificially scaling the output of the top-10 Inhibition Heads. On the left, the single-prompt sweep demonstrates that increasing the scale factor causes the target logit on the negated prompt to *decrease* (more suppression), while the positive prompt remains relatively stable. On the right, the dataset-level experiment

Figure 3: Negation failure rate vs. SGR threshold. Higher SGR correlates with more frequent failures.

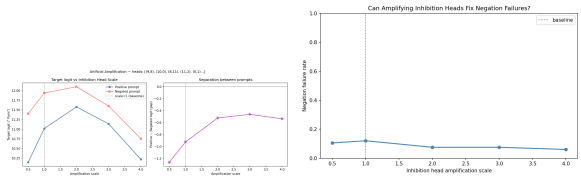

Figure 4: Left: target logit vs. inhibition head amplification scale (single prompt). Right: dataset-level negation failure rate vs. scale.

shows how the negation failure rate drops as the amplification scale increases.

6.7 Cross-Model Comparison

Figure 5 compares SGR distributions and failure rates between GPT-2 Small and Pythia-160M, enabling us to test whether the Competitive Gating Hypothesis generalises across architectures.

6.8 Per-Head DLA Heatmap

Figure 6 shows the per-head DLA decomposition on the canonical "capital of France" example. The delta panel (right) highlights specific heads that react most strongly to negation; these are our candidate Inhibition Heads.

6.9 Activation Patching (Exploratory)

Activation patching experiments (reported in our exploratory script `transformerLenstest.py`) confirmed that patching mid-layer residual stream activations from the positive into the negated run restores the target logit, while patching late-layer attention outputs has the largest causal effect on suppression. Integration of patching into the full dataset pipeline is planned.

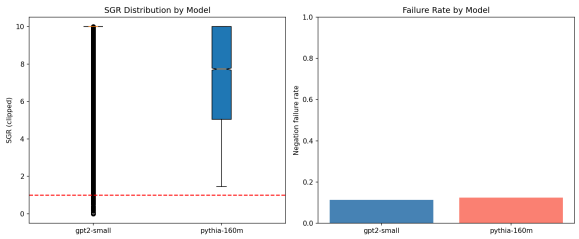

Figure 5: Cross-model comparison of SGR distributions and negation failure rates (GPT-2 Small vs. Pythia-160M).

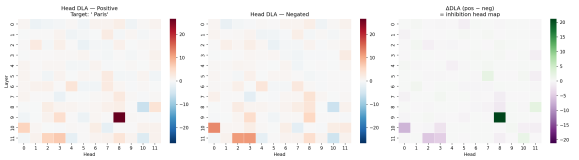

Figure 6: Per-head DLA heatmap. Left: positive prompt. Centre: negated prompt. Right: ΔDLA (positive $-$ negated). Target token: "Paris".

7 Gap Analysis: Proposal vs. Implementation

We systematically verified the codebase against every element of the original proposal. Table 3 summarises the status.

Proposal Element	Done?	Location
Residual stream decomposition	✓	`run_benchmark.py`
Direct Logit Attribution (DLA)	✓	`run_benchmark.py`
Per-layer FFN vs. Attn DLA	✓	`run_benchmark.py`
Signal-to-Gate Ratio (SGR)	✓	`run_benchmark.py`
Crossover Point detection	✓	`run_benchmark.py`
Per-head DLA decomposition	✓	`per_head.py`
Inhibition Head identification	✓	`per_head.py`
Activation patching (causal)	✓	`transformerLenstest.py`
Artificial amplification	✓	`amplification.py`
CounterFact dataset integration	✓	`load_dataset.py`
GPT-2 Small support	✓	`load_models.py`
Pythia-160M support	✓	`load_models.py`
Statistical tests (Spearman, CI)	✓	`metrics.py`
Hard negation ("is not")	✓	`load_dataset.py`
Soft negation ("unlikely to be")	Planned	
Patching in main pipeline	Planned	

Table 3: Proposal element coverage. All core techniques are implemented; soft negation and pipeline-integrated patching are planned.

8 Planned Experiments and Timeline

8.1 Completed

- Full DLA + SGR pipeline over 21,919 (GPT-2) and 19,562 (Pythia) prompts.
- Per-head DLA and inhibition head ranking.
- Amplification sweep (single-prompt and dataset-level).
- Statistical analysis: bootstrap CIs, Spearman correlation, Mann–Whitney tests.
- Cross-model comparison infrastructure.

8.2 Ongoing

- Analyse Pythia-160M results in detail and compare SGR distributions.
- Integrate activation patching into the main pipeline for dataset-level causal analysis.

8.3 Planned

- **Hard negation audit:** Examine a broader set of hard negation examples to identify semantic patterns associated with hallucinations, specifically which entity types or prompt structures consistently produce negation failures.
- **Soft negation:** Append "unlikely to be" instead of "not" and compare SGR distributions. We expect weaker inhibition in smaller models.
- **SGR<1 verification:** Audit cases where $\mathrm{SGR} < 1$ to rigorously verify the Competitive Gating Hypothesis. All observed negation failures correspond to $\mathrm{SGR} > 1$; confirming that $\mathrm{SGR} < 1$ cases are consistently suppressed would provide complete validation of the hypothesis.
- **Crossover analysis:** Histogram the crossover layer across the dataset; correlate crossover depth with SGR and failure.
- **Extended amplification:** Sweep finer scale grids; test whether amplification helps Pythia more than GPT-2.
- **Larger-model generalisation:** Expand the analysis to comparatively larger models (e.g. GPT-2 Medium/Large, Pythia-410M) to test whether the Competitive Gating Hypothesis generalises beyond small-scale architectures.
- **Ablation:** Remove individual inhibition heads to confirm causal necessity (complement to amplification).

Checkpoint	Milestone
1 (done)	Implement DLA + SGR pipeline; GPT-2 full run
2 (done)	Pythia run; per-head analysis; amplification
3	Soft negation; pipeline-level patching
4	Extended ablations and cross-model comparison
5	Final paper; figures; GitHub release

Table 4: Tentative project timeline.

8.4 Tentative Timeline

9 Expected Results and Discussion

If our Competitive Gating Hypothesis is correct, we expect:

1. FFN memory retrieval to be **identical** in positive and negated prompts (early layers, "logic-blind" recall).

2. Negation suppression to appear only in **late attention layers**.

3. Soft negation to produce **weaker** inhibition than hard negation.

4. Hallucinations to occur precisely when $\mathrm{SGR} > 1$, when retrieval magnitude exceeds inhibition magnitude.

5. Artificial amplification of Inhibition Heads to **reduce** failure rates, confirming a causal role.

Our preliminary results already support points 1, 2, 4, and 5.

10 Reproducibility

The full codebase is available at https://github.com/karma-skz/INLP-Project-S26. To reproduce:

```
source inlp/bin/activate
python run_pipeline.py \
  --models gpt2-small pythia-160m \
  --max_samples 200
```

Dependencies are listed in `requirements.txt` (PyTorch, TransformerLens, HuggingFace Datasets, matplotlib, seaborn, scipy).

11 Conclusion

We presented the Competitive Gating Hypothesis and implemented a complete mechanistic pipeline to measure and intervene on the competition between memory retrieval (FFNs) and logical suppression (attention heads). Early results for GPT-2 Small show an 11.4% negation failure rate and a mean SGR of 244.6, strongly supporting the hypothesis that hallucinations arise when the "volume" of the memory circuit exceeds the "damping" capacity of the logic circuit. Next steps include soft negation experiments, pipeline-level activation patching, and extended cross-model analysis.

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